

Rectangular Beam Stress Calculation Notes

ANSYS Static Structural Hand Check

1 Purpose

This document builds the hand calculation from the smallest quantities and compares it with the ANSYS result for the rectangular cantilever beam model. The goal is not to replace the finite-element result, but to check that the simulated stress and deformation are in the expected range.

2 Known Model Values

The values below were extracted from the ANSYS Workbench/SpaceClaim model and MAPDL solve files. ANSYS uses MKS units internally.

Symbol	Meaning	Value
L	Beam length along the y direction	6 mm = 0.006 m
w	Width along the x direction	2 mm = 0.002 m
t	Thickness along the z direction	1.25 mm = 0.00125 m
F_x	End force in the $+x$ direction	100 N
E	Young's modulus for structural steel	200 GPa = 200×10^9 Pa
ν	Poisson's ratio	0.30

Boundary conditions: the $y = 0$ face is fixed, and the load is applied on the $y = L$ face.

3 Cross-Section Area

Start with the rectangular cross-section. Area is width times thickness:

$$A = wt \tag{1}$$

Substitute the model dimensions:

$$A = (0.002)(0.00125) \tag{2}$$

$$= 2.50 \times 10^{-6} \text{ m}^2 \tag{3}$$

4 Average Applied Traction

The 100 N force is applied over the end face. The average surface traction is force divided by area:

$$p = \frac{F}{A} \tag{4}$$

$$p = \frac{100}{2.50 \times 10^{-6}} \quad (5)$$

$$= 4.00 \times 10^7 \text{ Pa} \quad (6)$$

$$= 40 \text{ MPa} \quad (7)$$

This matches the pressure magnitude written in the ANSYS solver input.

5 Cantilever Bending Moment

For a first hand check, treat the part as a cantilever beam with an end load. The maximum bending moment occurs at the fixed face:

$$M_{\max} = FL \quad (8)$$

$$M_{\max} = (100)(0.006) \quad (9)$$

$$= 0.600 \text{ N m} \quad (10)$$

6 Area Moment of Inertia

The beam axis is along y . The force is along x , so the beam bends about the z axis. For a rectangular section:

$$I_z = \frac{tw^3}{12} \quad (11)$$

$$I_z = \frac{(0.00125)(0.002)^3}{12} \quad (12)$$

$$= 8.33 \times 10^{-13} \text{ m}^4 \quad (13)$$

The farthest distance from the neutral axis is half the width:

$$c = \frac{w}{2} = \frac{0.002}{2} = 0.001 \text{ m} \quad (14)$$

7 Maximum Bending Stress

The basic bending stress equation is:

$$\sigma_b = \frac{Mc}{I} \quad (15)$$

Using the maximum moment and I_z :

$$\sigma_{b,\max} = \frac{M_{\max}c}{I_z} \quad (16)$$

$$= \frac{(0.600)(0.001)}{8.33 \times 10^{-13}} \quad (17)$$

$$= 7.20 \times 10^8 \text{ Pa} \quad (18)$$

$$= 720 \text{ MPa} \quad (19)$$

This is the simple beam-theory estimate. It assumes ideal beam behavior and does not fully capture the 3D stress concentration at the fixed face.

8 Tip Deflection Check

For a cantilever beam with an end load, the tip deflection is:

$$\delta_{\max} = \frac{FL^3}{3EI_z} \quad (20)$$

$$\delta_{\max} = \frac{(100)(0.006)^3}{3(200 \times 10^9)(8.33 \times 10^{-13})} \quad (21)$$

$$= 4.32 \times 10^{-5} \text{ m} \quad (22)$$

$$= 0.0432 \text{ mm} \quad (23)$$

9 Equivalent Stress

ANSYS equivalent stress is von Mises stress. For a general 3D stress state:

$$\sigma_{vm} = \sqrt{\frac{(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2 + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)}{2}} \quad (24)$$

For a mostly one-direction bending stress state, the von Mises stress is close to the bending stress magnitude:

$$\sigma_{vm} \approx |\sigma_b| \quad (25)$$

10 ANSYS Comparison

Quantity	Hand estimate	ANSYS result	Difference
Maximum equivalent stress	720 MPa	861.3 MPa	+19.6%
Estimated stress bound	720 MPa	917.7 MPa	+27.5%
Maximum deformation	0.0432 mm	0.0475 mm	+10.0%

Result locations from the MAPDL result file:

- Maximum deformation occurs near the free end, node 671, close to $(x, y, z) = (0, 6, 0.625)$ mm.
- Maximum equivalent stress occurs near the fixed-face corner region, including nodes 547 and 592.

The ANSYS stress is higher than the simple beam estimate because the part is short relative to its cross-section and the fixed support creates a local 3D stress concentration. The deformation comparison is closer because the global bending stiffness is captured reasonably well by the cantilever approximation.

11 Summary

The hand calculation predicts the correct order of magnitude for both deformation and stress. The final comparison supports the finite-element result: the model behaves like a very small cantilever beam, with maximum deformation at the free end and maximum equivalent stress near the fixed support.